cURL Commands - Complete Guide

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What is cURL?

cURL is a command-line tool used to transfer data to or from a server using various network protocols. It stands for "Client URL" and is a client-side tool to interact with URLs (Uniform Resource Locators).

Common Use Cases:

- Making HTTP or HTTPS requests
- Interacting with REST APIs
- Downloading/uploading files
- Sending data using different protocols (FTP, SMTP, etc.)

Version Check Commands

```bash
curl --version    # Check cURL version (Mac/Linux)
java -v           # Check Java version
java --version    # Alternative Java version check
```

Essential cURL Options

Option	Long Form	Description
-X	--request <command>	Specifies HTTP request method (GET, POST, PUT, DELETE, etc.)
-H	--header <header>	Adds custom header to the request
-d	--data <data>	Sends data in request body (POST/PUT requests)
-i	--include	Includes response headers in output
-o	--output <file>	Saves response body to a file
-L	--location	Follows redirects (301/302)
-u	--user <user:password>	Basic authentication credentials
-k	--insecure	Allows insecure SSL connections
-x	--proxy <[protocol://]host[:port]>	Specifies proxy for the request
-v	--verbose	Provides detailed debugging information
-m	--max-time <seconds>	Sets timeout for HTTP requests

Note: By default, cURL uses the GET method.

Important Points to Remember

1. URL Quoting

Always include URLs in single or double quotes when using multiple query parameters:

```bash
curl 'https://gorest.co.in/public/v2/users'
curl "https://gorest.co.in/public/v2/users"
```

2. Default Method

cURL uses GET method by default.

3. Timeout Settings

Use `-m` option to set timeout:

```bash
curl -m 1 'http://httpbin.org/get'
```

4. File Upload

Send files via HTTP POST using `@`:

```bash
curl --location 'http://httpbin.org/post' \
--form 'file=@"/Users/naveenautomationlabs/Desktop/hdfc.pdf"'
```

5. Proxy Usage

```bash
curl -x http://proxy.example.com:8080 https://api.ipify.org
curl -x http://51.158.68.26:8811 https://api.ipify.org
```

Authentication Methods

Basic Authentication

URL: https://the-internet.herokuapp.com/basic_auth

With credentials:

```bash
curl -i -u "admin:admin" -X GET "https://the-internet.herokuapp.com/basic_auth"
```

Prompt for password (more secure):

```bash
curl -i -u "admin" -X GET "https://the-internet.herokuapp.com/basic_auth"
```

OAuth 2.0 Token Request

```bash
curl --location 'https://test.api.amadeus.com/v1/security/oauth2/token' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--data-urlencode 'grant_type=client_credentials' \
--data-urlencode 'client_id=TAnRnsU5lASXZ8mPGdwRQZMoQzhu6Gwv' \
--data-urlencode 'client_secret=VjjgfcJilNAzcSJw'
```

REST API Examples

GET Request - List Users

```bash
```

```
curl -i \
  -H "Accept:application/json" \
  -H "Content-Type:application/json" \
  -H "Authorization: Bearer e4b8e1f593dc4a731a153c5ec8cc9b8bbb583ae964ce650a741113091b4e2ac6
  -X GET "https://gorest.co.in/public/v2/users"
```

POST Request - Create User

Linux/Mac/Git Bash:

```bash
curl -i \
  -H "Accept:application/json" \
  -H "Content-Type:application/json" \
  -H "Authorization: Bearer e4b8e1f593dc4a731a153c5ec8cc9b8bbb583ae964ce650a741113091b4e2ac6
  -X POST "https://gorest.co.in/public/v2/users" \
  -d '{"name":"Tenali Ramakrishna", "gender":"male", "email":"tenali.ramakrishna@15ce.com", "status":"ac
```

Windows Command Prompt:

```cmd
curl -i ^
  -H "Accept:application/json" ^
  -H "Content-Type:application/json" ^
  -H "Authorization: Bearer e4b8e1f593dc4a731a153c5ec8cc9b8bbb583ae964ce650a741113091b4e2ac6
  -X POST "https://gorest.co.in/public/v2/users" ^
  -d "{\"name\":\"Tenali Ramakrishna\", \"gender\":\"male\", \"email\":\"tenali.ramakrishna@15ce.com\", \"sta
```

PowerShell:

```powershell
curl -i `
  -H "Accept:application/json" `
  -H "Content-Type:application/json" `
  -H "Authorization: Bearer e4b8e1f593dc4a731a153c5ec8cc9b8bbb583ae964ce650a741113091b4e2ac6
  -X POST "https://gorest.co.in/public/v2/users" `
  -d '{"name":"Tenali Ramakrishna", "gender":"male", "email":"tenali.ramakrishna@15ce.com", "status":"ac
```

PATCH Request - Update User

```bash
curl -i \
  -H "Accept:application/json" \
  -H "Content-Type:application/json" \
  -H "Authorization: Bearer e4b8e1f593dc4a731a153c5ec8cc9b8bbb583ae964ce650a741113091b4e2ac6
  -X PATCH "https://gorest.co.in/public/v2/users/1932" \
  -d '{"name":"Allasani Peddana", "email":"allasani.peddana@15ce.com", "status":"active"}'
```

DELETE Request - Delete User

```bash
curl -i \
  -H "Accept:application/json" \
  -H "Content-Type:application/json" \
  -H "Authorization: Bearer e4b8e1f593dc4a731a153c5ec8cc9b8bbb583ae964ce650a741113091b4e2ac6
  -X DELETE "https://gorest.co.in/public/v2/users/1932"
```

Platform-Specific Line Continuation

Platform	Line Continuation Character
Linux/Mac/Git Bash	`\` (backslash)
Windows CMD	`^` (caret)
PowerShell	`` ` `` (backtick)

HTTP Redirection

- **301**: Moved permanently
- **302**: Found (temporary redirect)
- Use `-L` or `--location` to follow redirects automatically

Example flow: Client → Server 1 → Server 2 → Server 3 → Final Response

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Security Best Practices

1. **Avoid plaintext passwords in shell history** - Use the password prompt method
2. **Use HTTPS when possible** - Especially for authentication
3. **Be cautious with `-k` flag** - Only use for testing with self-signed certificates
4. **Store tokens securely** - Consider environment variables for sensitive data

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These notes cover the essential cURL commands and examples for API testing and automation.

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